Project Initialization and Planning Phase

Date	28 June 2026
Team ID	xxxxxx
Project Title	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The **Emotion Detection and Learning Support Engine** is an Artificial Intelligence-based system developed to recognize users' emotions from text input and provide personalized learning support. The project aims to improve students' learning experience by detecting emotions such as happiness, sadness, anger, fear, surprise, and neutrality using a **BERT-based Natural Language Processing (NLP)** model. Based on the detected emotion, the system generates appropriate study recommendations, motivational messages, and learning resources through an interactive **Streamlit** web application. This intelligent approach enhances student engagement, emotional well-being, and academic performance by delivering personalized assistance.

Project Overview	
Objective	The primary objective of the project is to develop an AI-powered emotion detection system that accurately identifies users' emotions from text input and provides personalized learning recommendations to improve engagement, motivation, and academic performance.
Scope	The project focuses on emotion recognition using Natural Language Processing (NLP), integrating a BERT-based deep learning model with a Streamlit web application. The system analyzes user text, predicts emotions, and provides customized learning support in real time.
Problem statement	

Description	Students often experience stress, anxiety, frustration, and low motivation during learning. Existing online learning platforms mainly focus on delivering educational content and rarely consider the emotional state of learners, resulting in reduced engagement and poor learning outcomes.
Impact	Addressing these challenges will improve students' emotional well-being, increase learning motivation, provide personalized study support, enhance academic performance, and create a more effective and engaging learning environment.
Proposed Solution	
Approach	Develop an AI-powered system using a **BERT-based NLP model** to analyze user text, detect emotions, and generate personalized learning recommendations through a Streamlit web application.
Key Features	• AI-based emotion detection using BERT. • Personalized learning recommendations based on detected emotions. • Motivational messages and study guidance.
	• Interactive Streamlit web interface. • Real-time emotion prediction and learning support. • Accurate and efficient NLP-based emotion classification.

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	CPU/GPU specifications	Intel Core i5/i7 Processor or Google Colab with NVIDIA T4 GPU
Memory	RAM specifications	Minimum 8 GB RAM

Storage	Space for datasets, trained models, and project files	256 GB SSD or higher
Software		
Frameworks	Python frameworks	Streamlit
Libraries	Python Libraries	Transformers, PyTorch, Pandas, NumPy, Scikit-learn, Matplotlib, Joblib
Development Environment	IDE	Visual Studio Code (VS Code), Jupyter Notebook, Google Colab
Data		
Dataset	Source, size, format	**Emotion Dataset for NLP (CSV format)** obtained from **Kaggle/Hugging Face**, containing thousands of labeled text samples representing emotions such as joy, sadness, anger, fear, surprise, and neutral.